

Part: **3.8 – COMPOSITE ASSEMBLIES**

Chapter: **3850**

Title: **COMPOSITE SPARS**

Revision: **0.1**

Date: **08/05/2021**

1 INTRODUCTION

1.1 METHOD-SUMMARY

1.2 SCOPE AND LIMITATIONS

2 STRUCTURAL DESCRIPTION

3 LOADS

3.1 WEB IN-PLANE LOADS

Standard simplification for a spar bay (between ribs stations or vertical stiffeners):

- ✓ X-axial load: linear distribution between lower and upper caps
- ✓ Y-axial load: uniform
- ✓ In-plane shear uniform

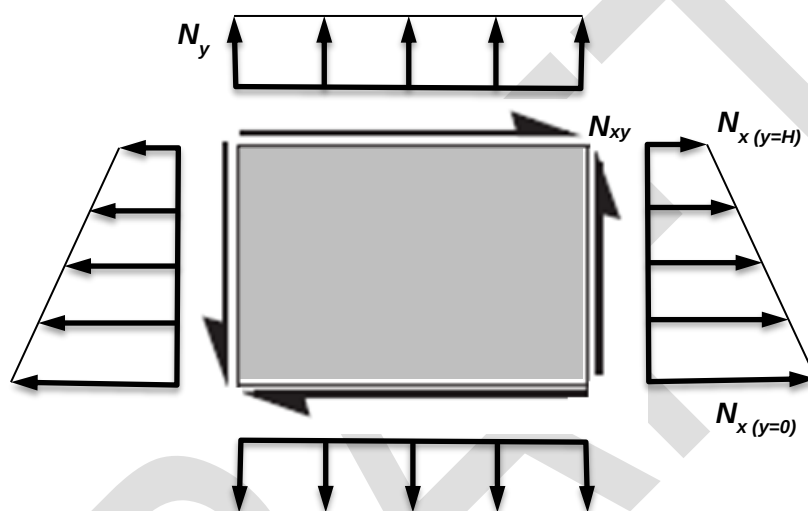


Figure 1. In plane internal loads in a web sub-panel

Loads extraction from GFEM

Y-axial and in-plane shear loads (N_y , N_{xy}) taken directly from 2D elements modelling the spar web, averaged in height in case several elements in height were used for the spar web.

X load shall be preferably taken from lower and upper spar caps, taking the longitudinal axial strain (ϵ_x) at those locations and converting it into web forces simply by imposing strain compatibility:

$$N_x = (E_x \cdot t \cdot \epsilon_x)_{web}$$

Pressure loads

4 STRENGTH CHECK METHODS

4.1 SPAR CAPS

4.1.1 Residual strength after BVID

Chapter 3405 "RESIDUAL STRENGTH AFTER TRANSVERSE BVID IMPACT"

Maximum compression longitudinal strain compared against the minimum CAI value of the spar segments should result in the critical margin for this failure mode.

4.1.2 Crippling

Chapter 3430 "LOCAL BUCKLING AND CRIPPLING OF BEAMS/STRINGERS"

4.1.3 Plain strength

Covered by previous strength checks if classified as damage tolerance structure.

Chapter 3403 "PLAIN STRENGTH OF COMPOSITE LAMINATES"

4.2 WEB PANEL

4.2.1 Residual strength after BVID

Chapter 3405 "RESIDUAL STRENGTH AFTER TRANSVERSE BVID IMPACT"

Holed webs

- ✓ DT-strength at web

For doing this calculation, amplify gross in-plane loads within the web (using both N_{xlo}, N_y, N_{xy} loads at lower and N_{xup}, N_y, N_{xy} at upper end), with the net-section reduction factor:

$$(N_{xlo}, N_y, N_{xy})_{dt} = f \cdot (N_{xlo}, N_y, N_{xy})_{web-gross} \quad \dots \text{where:} \quad f = H / (H - D)$$

- ✓ DT-strength at hole edge

Compare the peak compression strain at hole edge (see chapter 3404 "STRENGTH IN NOTCHED LAMINATES") with the applicable CAI design value for that detail.

4.2.2 Plain strength

Covered by previous strength checks if classified as damage tolerance structure.

Holed webs

Chapter 3404 "STRENGTH IN NOTCHED LAMINATES"

4.2.3 Web buckling

Chapter 3450 "PLATE/SHELL BUCKLING"

Buckling correction for holed webs

Chapter 3465 "HOLED PLATES AND WEBS"

Apply the following correction to the buckling load calculated for the web as if it did not include holes or notches:

$$KDF = 1.0 - 1.4 \cdot D / \min(L, H)$$

4.2.4 Folded angles

The web panel may incorporate folded flanges (if not fastened L or T profiles) which permit the installation of fasteners at its joints with other structural members along its perimeter, eg. joints to skin panels or ribs.

In particular pressure load cases can induce tension loads in those fasteners, interlaminar shear and unfolding through-the-thickness stresses at those folded details. Check the margin for the delamination failure acc. to chapter 3495 "DELAMINATION/UNFOLDING".

4.3 STIFFENERS/STABILIZERS

4.4 FASTENED JOINTS

Joints to skinJoints to ribs

5 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
BVID	-	Barely Visual Impact Damage
CAI	$\mu\epsilon$	Residual compression strength after BVID
D	mm	Diameter of hole
H	mm	Height of web
KDF	-	Knock down factor
L	mm	Distance between two vertical stiffening member (e.g. ribs junctions)
N	N/mm	In-plane loads
X,Y	-	Local reference axis system at spar web, with X longitudinal direction parallel to wing span and Y vertical from lower to upper caps

REFERENCES

<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1 TAE-XX-XX-XXXXXX	X	This is the reference number one (style "Reference"). In this case, an especially long document title is used.

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 1. Software programs referenced in this chapter

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.1 DD/05/2021	G. Baños	First issue	All pages

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Methods (TEPAP-TL4)	G. Baños DD/MM/AAAA	F. Glock DD/MM/AAAA

Table 3. Approval signatures table for last revision of this chapter